Superstore Sales Analysis - Executive Summary

ApexPlanet Data Analytics Internship - Task 5 Final Report

Executive Summary

Over the four-year period from January 2014 to December 2017 the Superstore recorded $2,297,200.86 in total sales and $286,397.02 in profit across 5,009 orders placed by 793 customers, a 12.47% profit margin. The average order was worth $458.61 and carried a 15.62% discount, and profitability turns negative once discounts pass the 20% mark. Technology is the strongest category by profit ($145,454.95), the West is the largest region by sales ($725,457.82) while the Central region is the weakest on margin at 7.92%. Demand is strongly seasonal, peaking every November and December, with November 2017 alone reaching $118,447.83. The results point to three practical levers: disciplined discounting, seasonal planning and segment-specific marketing.

Key Performance Indicators

KPI	Value
Total Sales	$2,297,200.86
Total Profit	$286,397.02
Profit Margin	12.47%
Orders	5,009
Customers	793
Products	1,862
Average Order Value	$458.61
Average Discount	15.62%

Supporting Charts

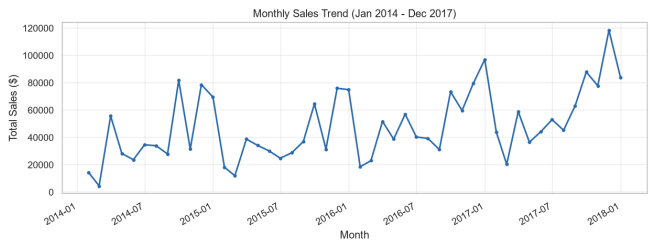


Figure 1 - Monthly sales trend (2014-2017)

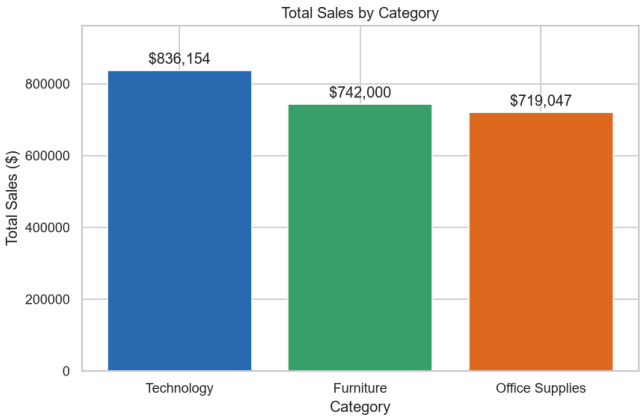


Figure 2 - Total sales by category

Insights, Recommendations & Next Steps

Top 5 Insights

1. Seasonality is strong. Sales peak every November and December; November 2017 hit $118,447.83 and the seasonal strength score is 0.805 (trend strength 0.279), so demand is far from flat across the year.

2. Discounting destroys margin. Profit turns negative above a 20% discount and at 41%+ the margin is -77.40%, losing $99,558.59 in total; discounted lines average -$6.66 profit versus +$66.90 at full price (p ~ 4.4e-55).

3. Technology is the profit engine. Technology delivered $145,454.95 of profit, while Furniture's $741,999.80 of sales produced only $18,451.27 - high revenue with thin return.

4. Customer value is concentrated. 264 Champions (33.3% of customers) spend $5,024.46 each versus $1,645.11 for Discount seekers; even the top spender, Sean Miller, is unprofitable at -$1,980.74.

5. Growth and margin live in different regions. The West is the largest region by sales ($725,457.82) while the Central region is the weakest on margin at just 7.92%, so a single national playbook misallocates effort.

Business Recommendations

1. Cap routine discounts and reprice the 21%+ bands. Reserve deep discounts for genuine clearance only, since every point of discount past 20% erodes margin directly (41%+ loses money at a -77.40% margin).

2. Plan stock, staffing and targets around the Nov/Dec peak. Budget against the seasonal curve instead of a flat run-rate; this also sharpens forecasting, which is currently limited (3-month moving-average backtest MAPE 50.58%).

3. Run segment-specific plays. Loyalty rewards for Champions, cross-sell for the Steady core (34.4% of customers), bundle pricing instead of percentage-off for Discount seekers, and one low-cost reactivation email for the 88 Dormant customers.